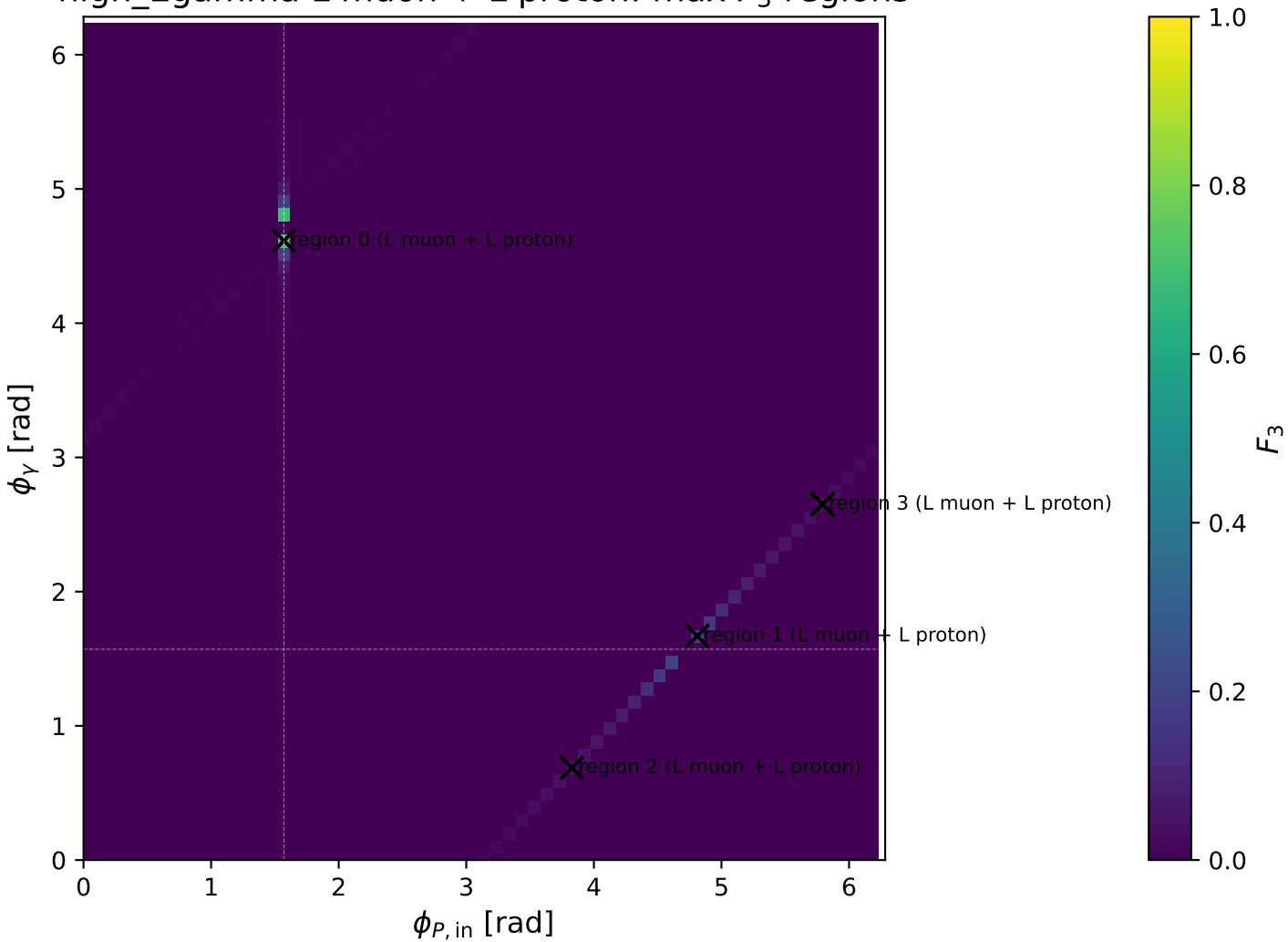
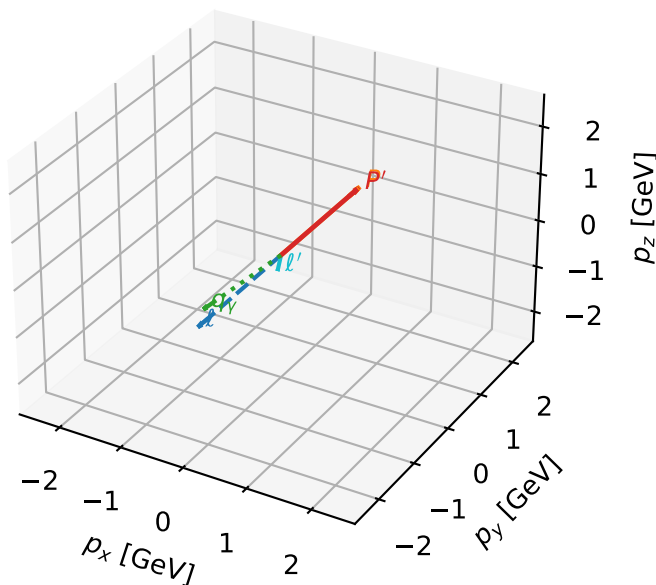


high_Egamma L muon + L proton: max F_3 regions



L muon + L proton characteristic max F_3 configuration

3D momenta

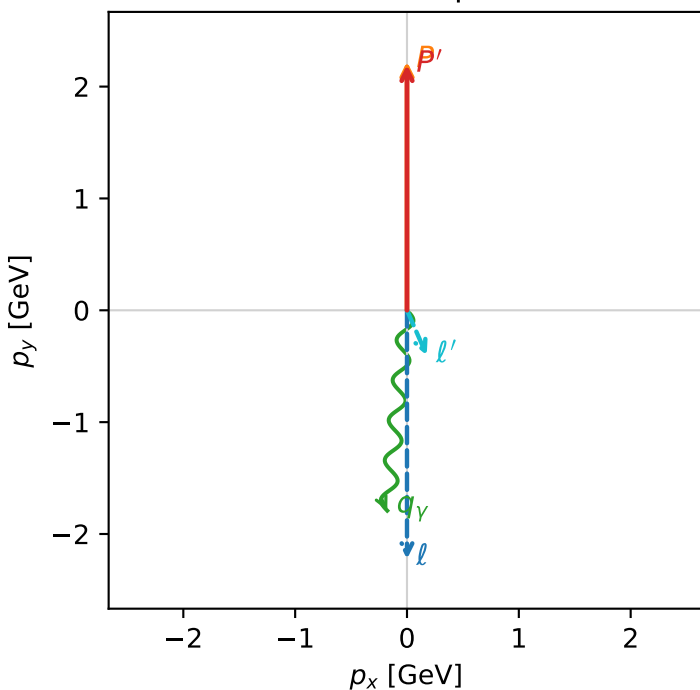


f3_high_egamma_ll_region_0 F_3=0.685592
 region: high_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

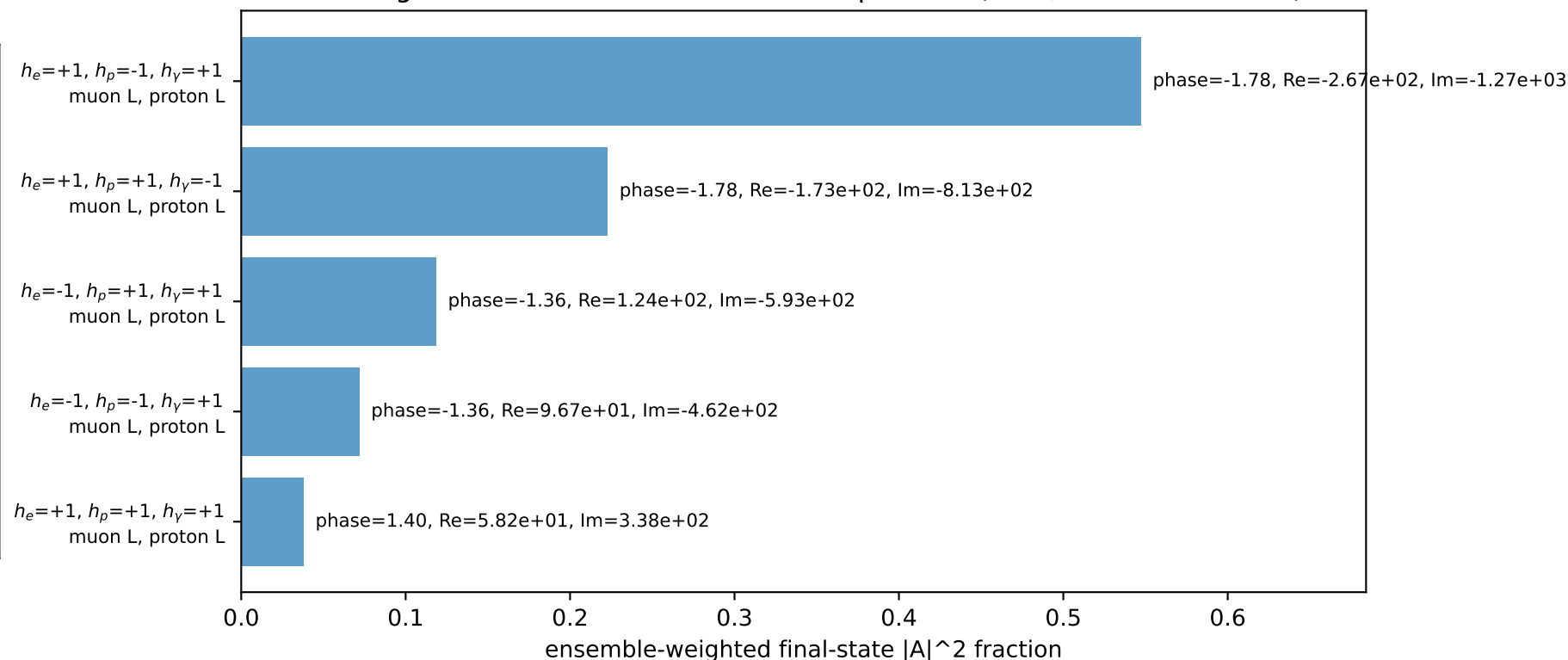
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.19415$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_p=1.5708$
 $E_Y=1.8$, $\phi_Y=4.61421$
 $Q^2=0.199845$, $x_B=0.0120018$, $t=-0.000128163$, $W^2=17.3312$, $y=0.807338$
 $\theta(l', \gamma)=29.278$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= 0.0000$
 l' $E= 0.4523$ $p_x= 0.1764$ $p_y= -0.4028$ $p_z= 0.0000$
 P' $E= 2.3862$ $p_x= 0.0000$ $p_y= 2.1942$ $p_z= 0.0000$
 q_Y $E= 1.8000$ $p_x= -0.1764$ $p_y= -1.7913$ $p_z= 0.0000$

Transverse plane

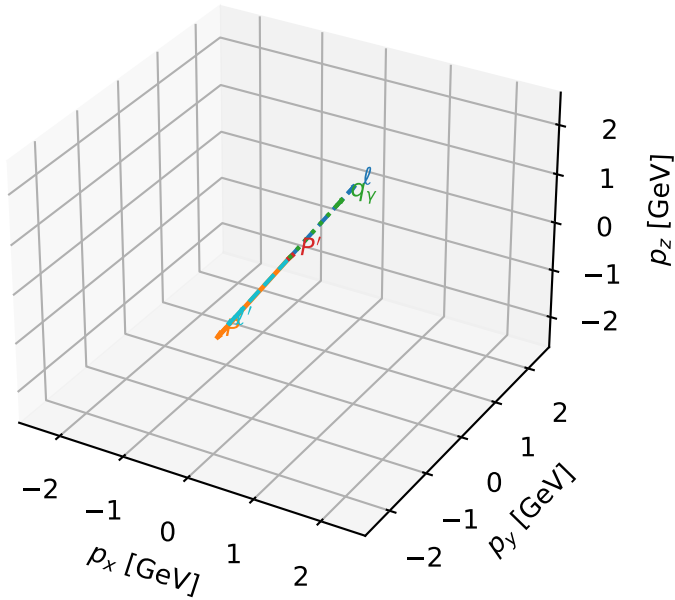


Leading initial-ensemble/final-state components (N=5, retained=99.9%)



L muon + L proton characteristic max F_3 configuration

3D momenta

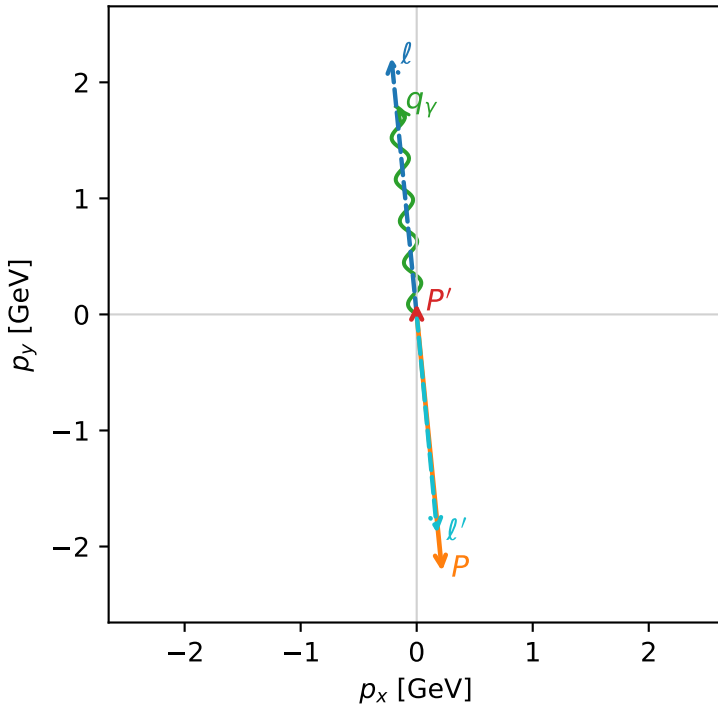


f3_high_egamma_ll_region_1 F_3=0.195276
 region: high_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

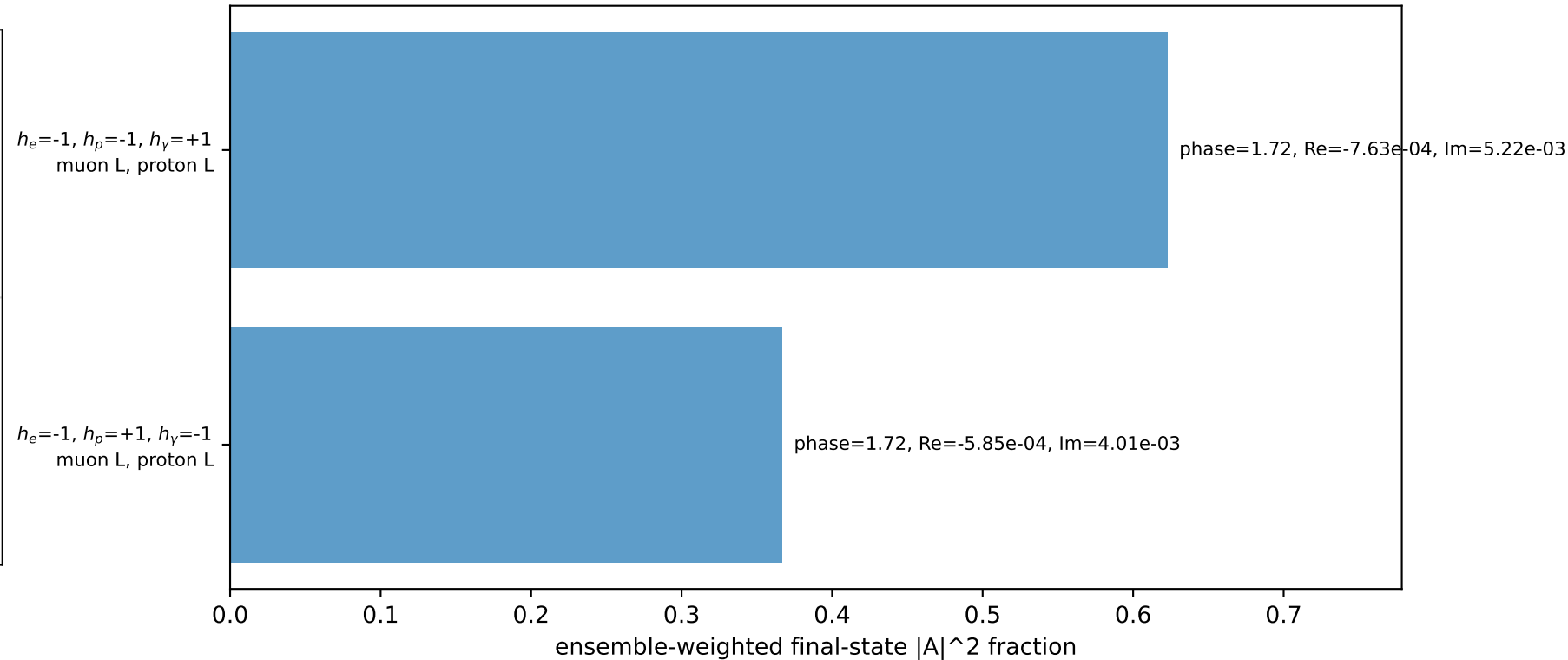
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.0933634$
 $\theta_{in}=1.57079$, $\phi_l=1.66897$, $\phi_p=4.81056$
 $E_\gamma=1.8$, $\phi_\gamma=1.66897$
 $Q^2=16.833$, $x_B=0.84622$, $t=-3.20239$, $W^2=3.93884$, $y=0.964469$
 $\theta(l', \gamma)=179.723$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.2179$ $p_y= 2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.2179$ $p_y= -2.2124$ $p_z= 0.0000$
 l' $E= 1.8959$ $p_x= 0.1764$ $p_y= -1.8847$ $p_z= 0.0000$
 P' $E= 0.9426$ $p_x= 0.0000$ $p_y= 0.0934$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -0.1764$ $p_y= 1.7913$ $p_z= 0.0000$

Transverse plane

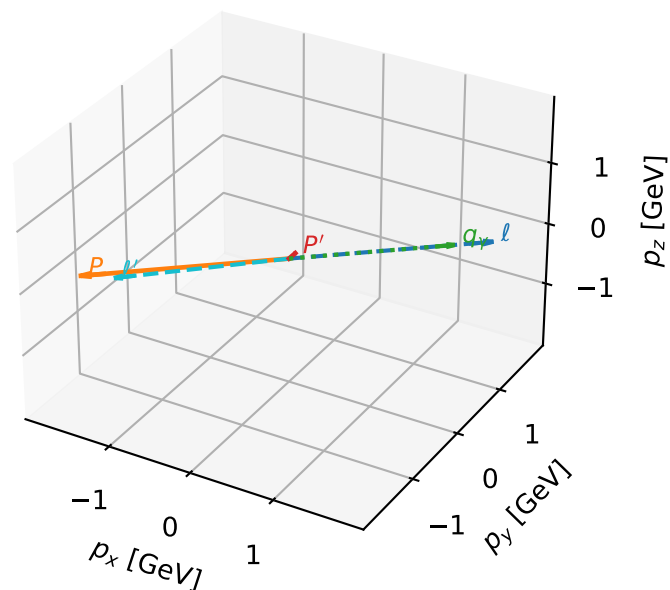


Leading initial-ensemble/final-state components (N=2, retained=99.0%)



L muon + L proton characteristic max F_3 configuration

3D momenta

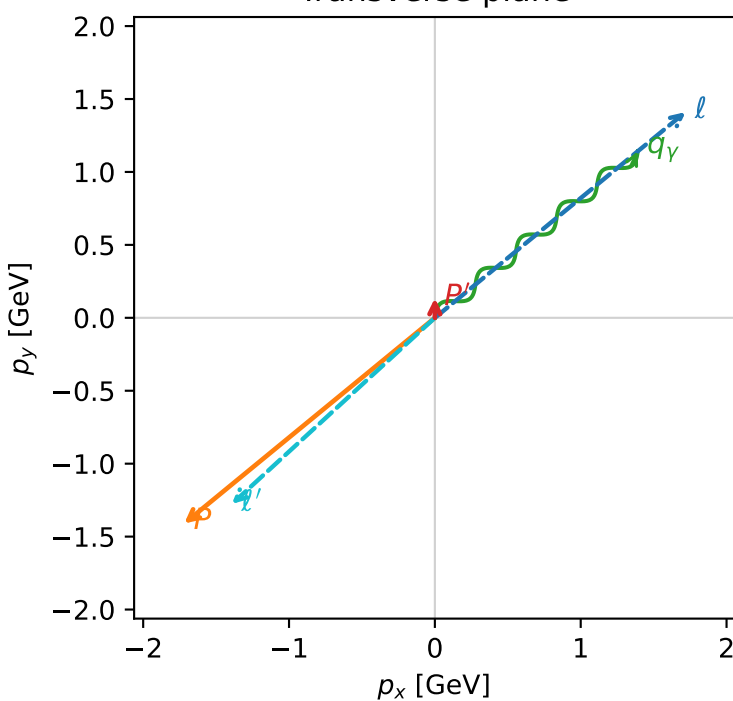


f3_high_egamma_ll_region_2 F_3=0.0475577
 region: high_Egamma_LL_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.13423$
 $\theta_{in}=1.57079$, $\phi_l=0.687223$, $\phi_p=3.82882$
 $E_\gamma=1.8$, $\phi_\gamma=0.687223$
 $Q^2=16.7766$, $x_B=0.843841$, $t=-3.19163$, $W^2=3.98449$, $y=0.963948$
 $\theta(l', \gamma)=176.85$ deg

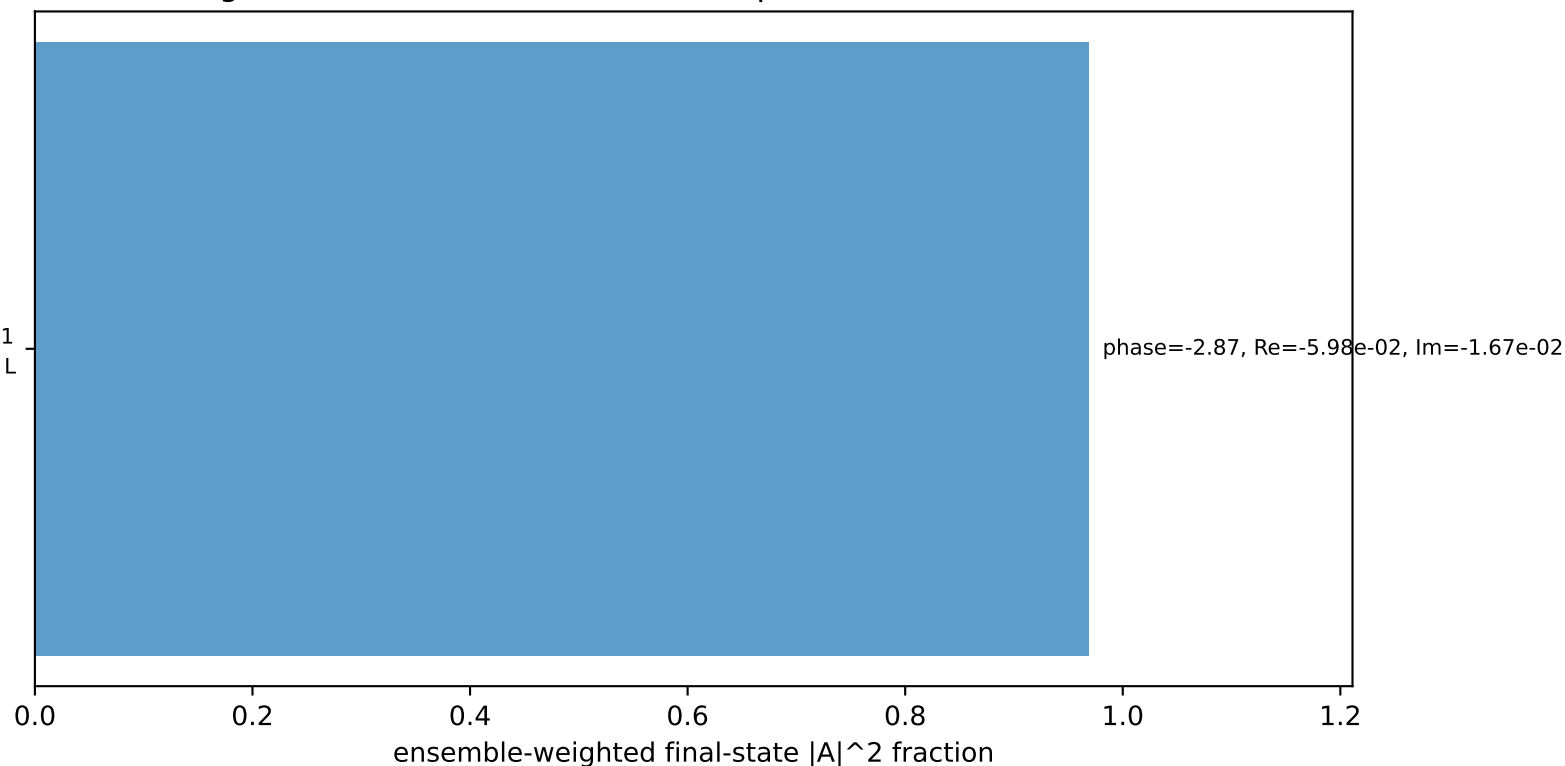
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.7185$ $p_y= 1.4103$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.7185$ $p_y= -1.4103$ $p_z= 0.0000$
 l' $E= 1.8910$ $p_x= -1.3914$ $p_y= -1.2761$ $p_z= 0.0000$
 P' $E= 0.9476$ $p_x= 0.0000$ $p_y= 0.1342$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= 1.3914$ $p_y= 1.1419$ $p_z= 0.0000$

Transverse plane



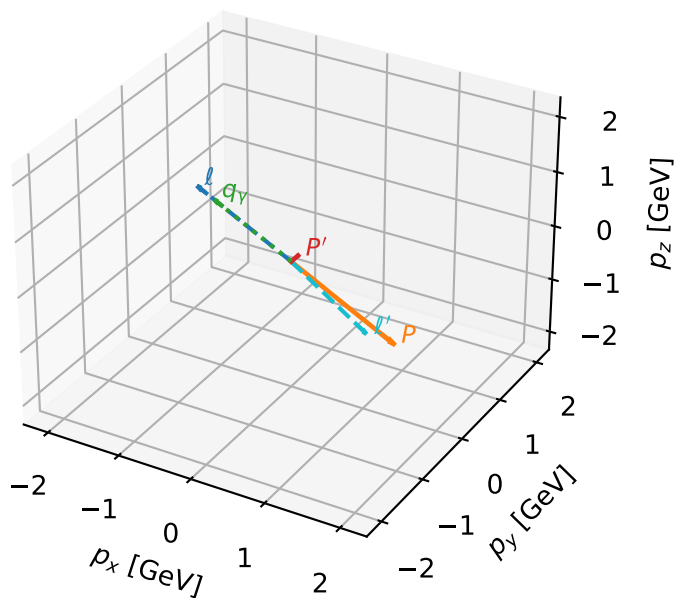
$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
 muon L, proton L

Leading initial-ensemble/final-state components (N=1, retained=96.9%)



L muon + L proton characteristic max F_3 configuration

3D momenta



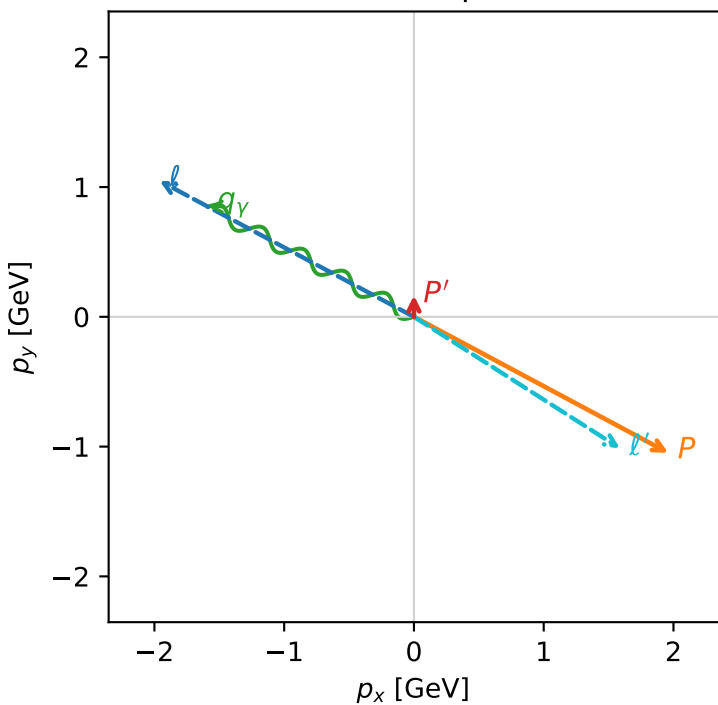
f3_high_egamma_ll_region_3 F_3=0.0388992
region: high_Egamma_LL_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.16464$
 $\theta_{in}=1.57079$, $\phi_l=2.65072$, $\phi_p=5.79231$
 $E_\gamma=1.8$, $\phi_\gamma=2.65072$
 $Q^2=16.7218$, $x_B=0.841525$, $t=-3.18118$, $W^2=4.02887$, $y=0.963441$
 $\theta(l', \gamma)=175.578$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l	$E=$	2.2256	$p_x=$	-1.9606	$p_y=$	1.0480	$p_z=$	-0.0000
P	$E=$	2.4129	$p_x=$	1.9606	$p_y=$	-1.0480	$p_z=$	0.0000
l'	$E=$	1.8862	$p_x=$	1.5875	$p_y=$	-1.0132	$p_z=$	0.0000
P'	$E=$	0.9523	$p_x=$	0.0000	$p_y=$	0.1646	$p_z=$	0.0000
q_γ	$E=$	1.8000	$p_x=$	-1.5875	$p_y=$	0.8485	$p_z=$	0.0000

Transverse plane



$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
muon L, proton L

Leading initial-ensemble/final-state components (N=1, retained=97.7%)

